*Preliminaries*

# From the Molecule to the cell-I

## Important issues

-Lectures

-Team-based learning (TBL)

-Workshops

**Microscopy Lab**
***(Ed. Ampliación de Bibliotecas)***




**Previous to the TBL:**
-Study the material indicated in the homework guide

**During the TBL:**

1-Quiz (test), individually.

2-Same test, in groups.

3-Clarification of some topics.

4-Practical case/clinical case.

**Quizzes with ADI: bring your phone, Laptop or tablet**

# Materials to read previous to the TBL

## STUDY MATERIAL LEC.1.1.

**SUMMARY AND TRANSLATION OF CHAPTER 1, FROM THE BOOK "BIOLOGÍA CELULAR BIOMÉDICA", INTRODUCCIÓN A LA BIOLOGÍA DE LAS CÉLULAS**

Authors: Alfonso Calvo, Carlos Gamazo

(Translation by Alberto Pezonaga)

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**Learning objectives:**

- **To know the general characteristics of eukaryotic cells.**
- **To define the specific features of prokaryotic cells, comparing them to those of eukaryotic ones.**
- **To know the basic organization of subcellular agents such as viruses, viroids and prions.**

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### INTRODUCTION

Cell Biology approaches the study of cells from a comprehensive perspective including morphological, the structural, functional and reproductive units) of all living things. In 1885, pathologist Rudolph Virchow completed this doctrine by stating that "all cells

## MATERIAL DE ESTUDIO LEC.1.1.

**MATERIAL RESUMIDO DE BIOLOGÍA CELULAR BIOMÉDICA, CAPÍTULO 1: INTRODUCCIÓN A LA BIOLOGÍA DE LAS CÉLULAS**

Autores: Alfonso Calvo, Carlos Gamazo

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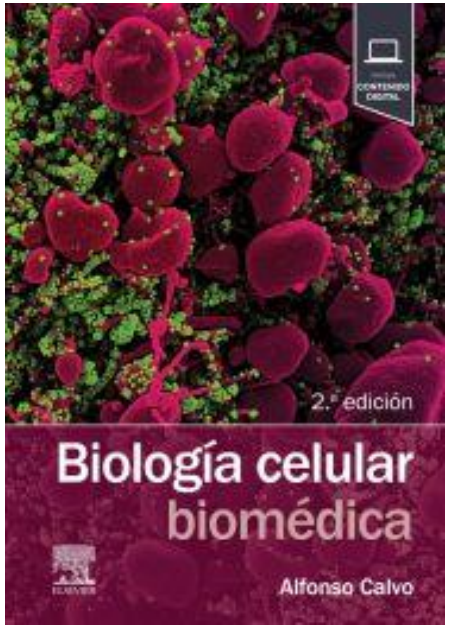
**Objetivos de aprendizaje:**

- **Conocer las características generales de las células eucariotas.**
- **Definir los rasgos propios de las células procariotas, comparándolos con los de las células eucariotas.**
- **Conocer los elementos básicos de organización de microorganismos acelulares, como los virus, viroides y priones.**

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### INTRODUCCIÓN

bacterias, a los que denominó animáculos. En 1837-1839, Schleiden y Schwann definieron la teoría celular



# Materials to study for the first TBL (Thursday 3rd 2026)

**1-Study Material corresponding to Lec 1.1. (on ADI) and notes from the class**

**2-Short videos from Lec 1.1. on ADI**

**3-Document on ADI (folder TBL 1.1.) on Ultrastructure of the eukaryotic cell.** Goal: to recognize the cell organelles

# General organization of cells

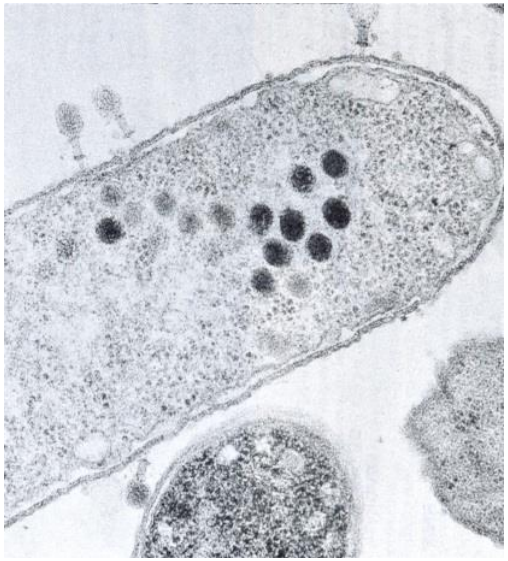
## Overview

-Types of cells. Prokaryotic vs. Eukaryotic cells

-General organization of  bacteria

-Sub-cellular agents: viruses

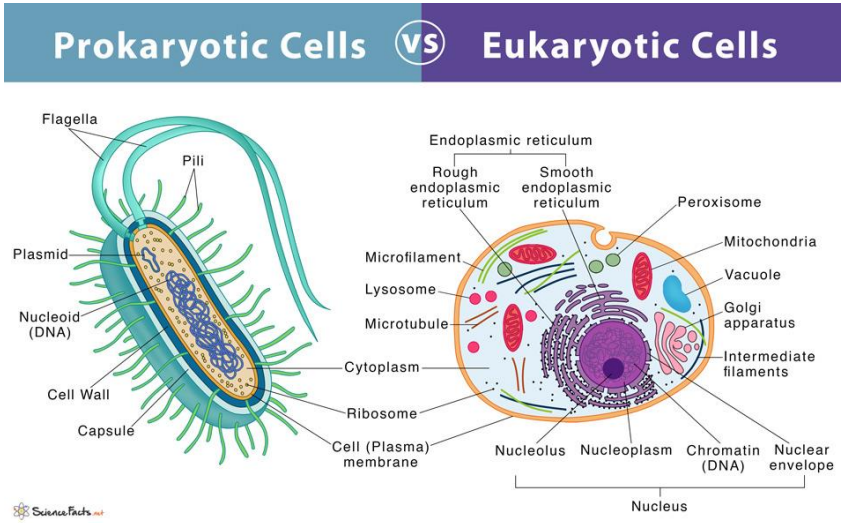
-Taking advantage of viruses: gene therapy



Bacteria and bacteriophages

# Prokaryotic versus eukaryotic cells

| Prokaryotic | Eukaryotic |
|---|---|
| Small cells (< 5um) | large cells (> 10um) |
| always unicellular | often multi cellular |
| no nucleus instead has a 'naked loop' called a nucleoid | always has nucleus with linear DNA (chromosomes) and histones |
| no membrane bound organelles | membrane bound organelles |
| ribosomes are small (70s) s = **svedberg** unit of measure of the size of organelles. | ribosomes are large (80s) |
| no mitochondria | mitochondria present |
| cell division by binary fission | cell division by mitosis or meiosis |
| reproduction mainly asexual however there are some non meiotic forms of gene exchange such as **conjugation**. | reproduction asexual or sexual |
| many metabolic pathways, fermentation, nitrogen fixation and photosynthesis. | common form of respiration involving organic molecules like sugar. |




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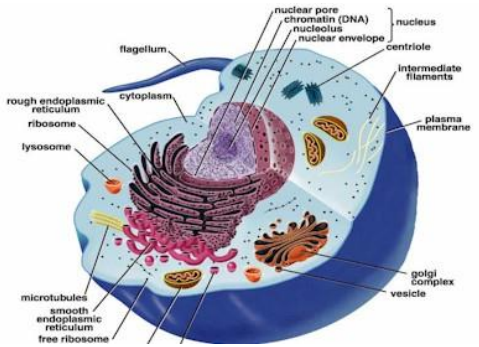
INTRODUCTION

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Review text on ADI corresponding to Lec.1.1. for more details

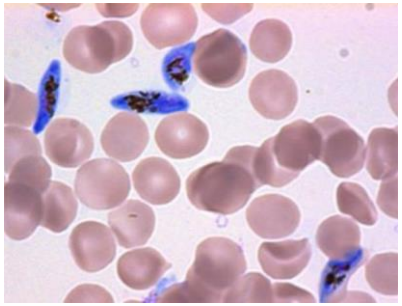
# Eukaryotic cells: human vs. other parasitic organisms

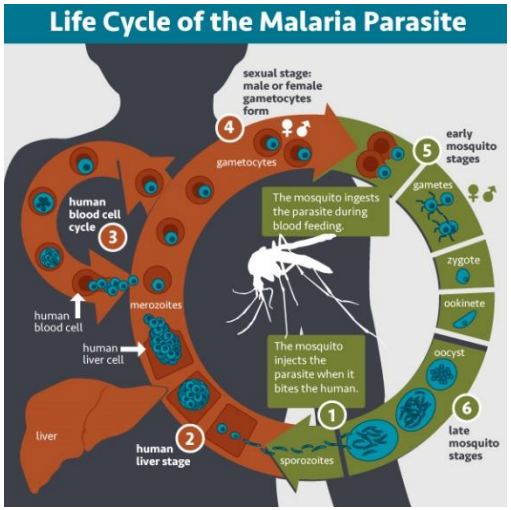
-Unicellular eukaryotes
-Free-living or parasites
-Complex life cycles
-Asexual/sexual reproduction



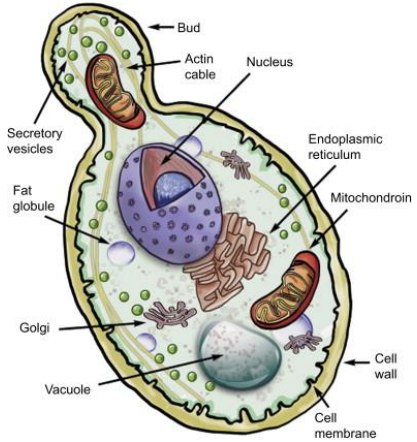

-Unicellular/multicellular eukaryotes
-Free-living or parasites
-Polysaccharide wall
-Asexual/sexual reproduction

***PROTOZOA:***
*Plasmodium falciparum*





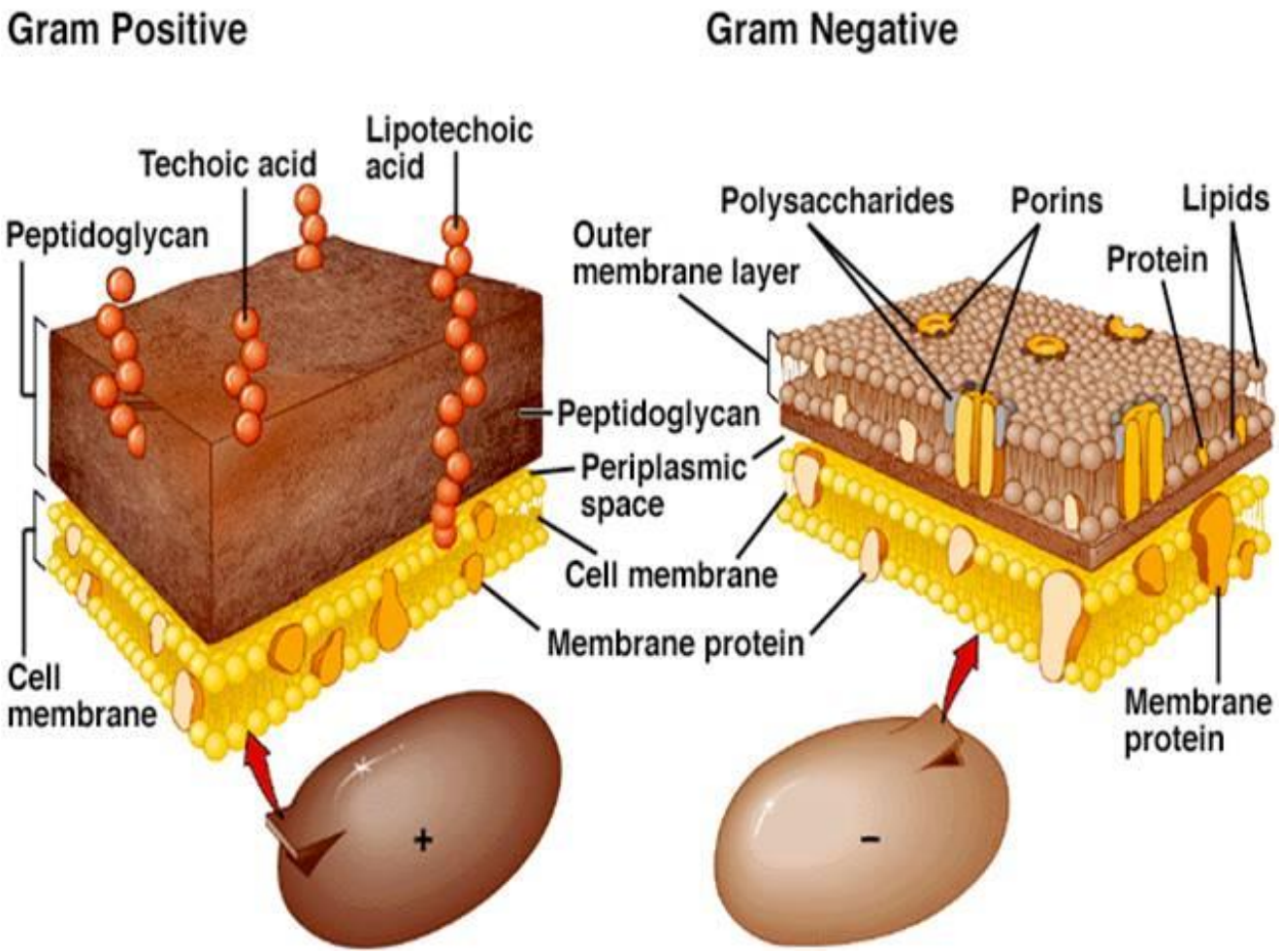

***FUNGI:*** *Candida albicans*

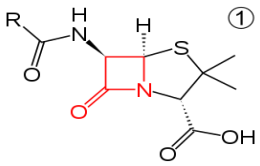






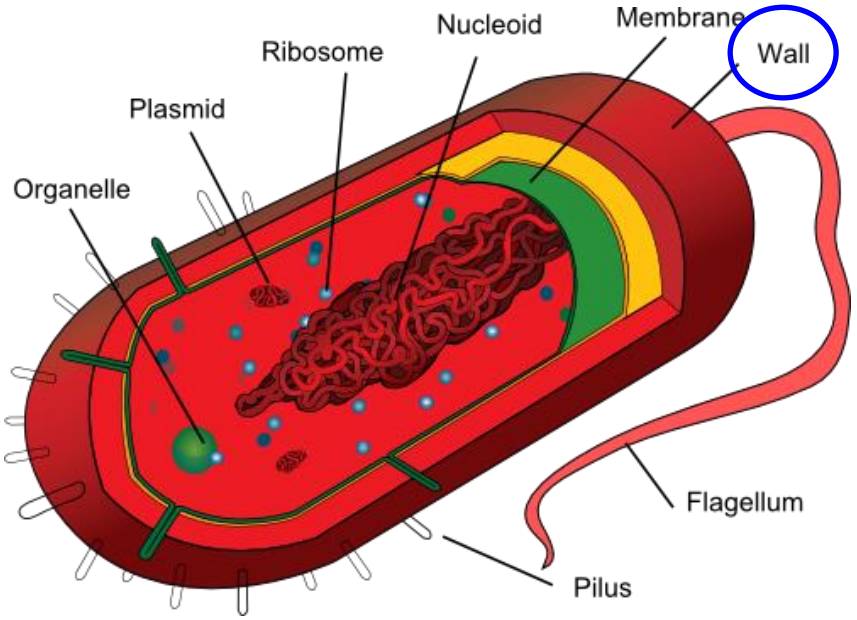

# Prokaryotic organisms: focus on bacteria. Gram positive vs. Gram negative



Penicillin (beta-lactam ring)

# Basic organization of bacteria




**MEMBRANE APPENDAGES**

-Flagellum

-Pili

-Fimbriae

-Ribosomes 70S

-Nucleoid: chromosome

## Plasmids:

-Extrachromosomal DNA

-Transferred through conjugation

-Survival advantage (antibiotic resistance)







-Used in genetic engineering to produce ***recombinant proteins***: e.g. **insulin**

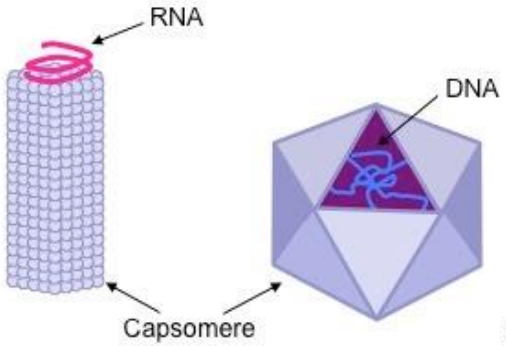
Watch the video on production of Recombinant insulin on ADI

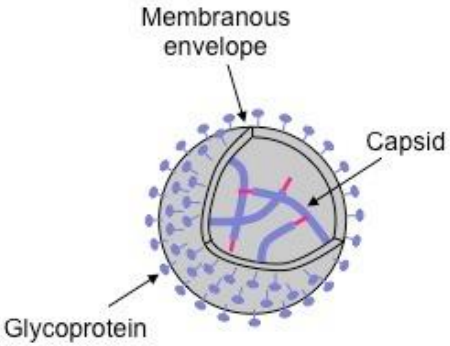
# Viruses

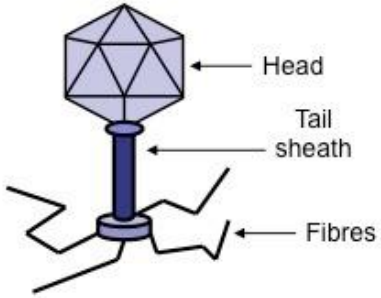
**Viruses**

30-100 nm

- **Nucleic acid** DNA/ RNA, linear or circular, single or double stranded
- **Capsid** Shell of proteins
- **Envelope** Lipid membrane

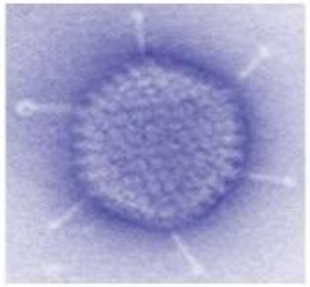


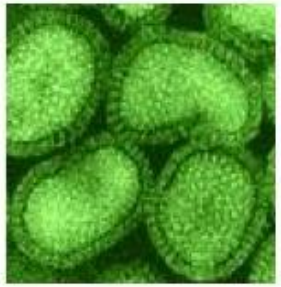





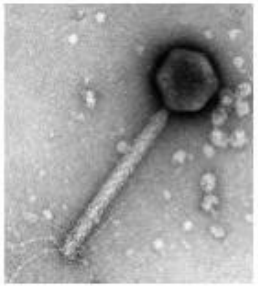

**Tobacco Mosaic Virus**



**Adenovirus**



**Influenza Virus**



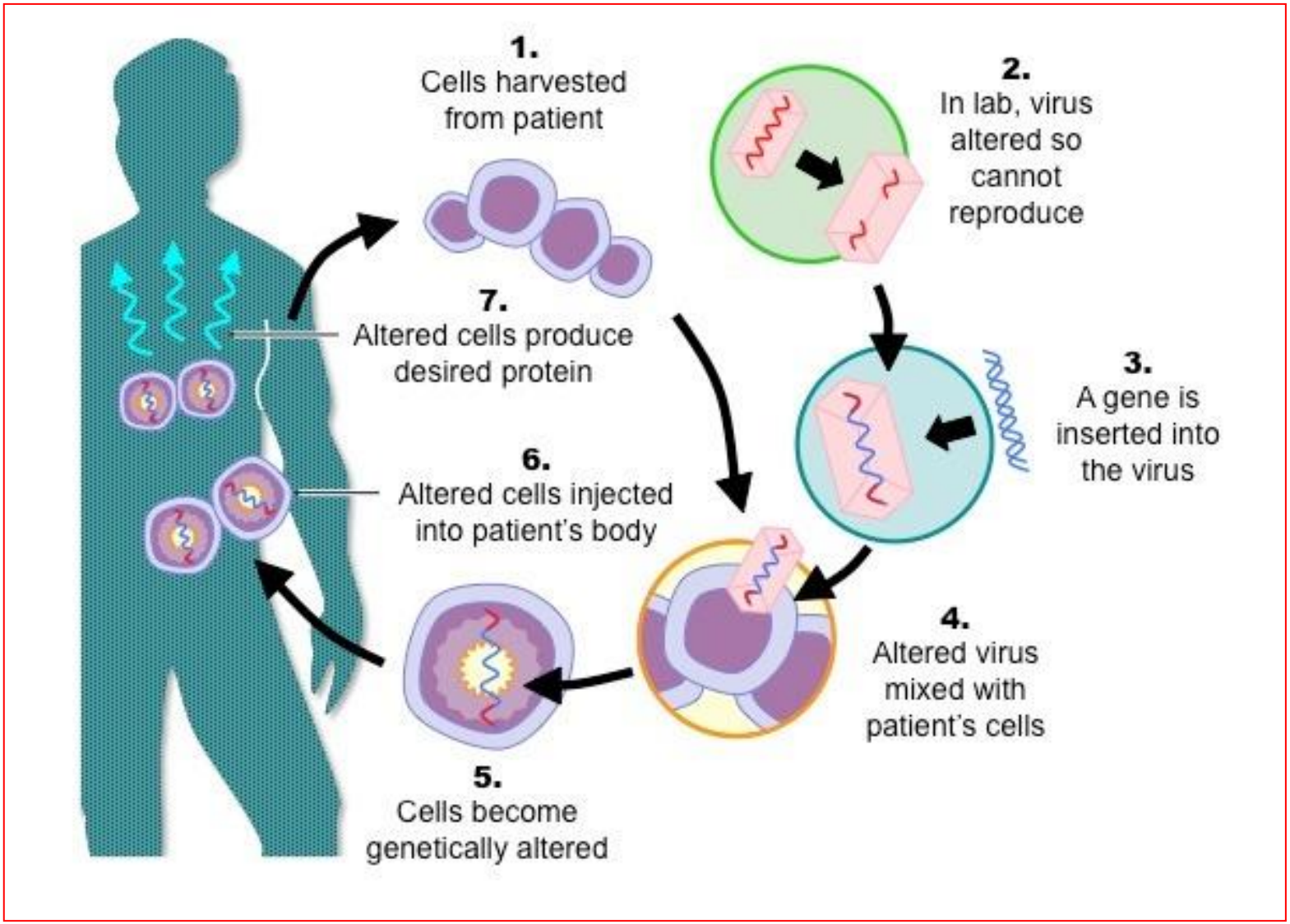
**Bacteriophage**

Bacteriophages



Attackers being attacked…

# Taking advantage of viruses: Gene therapy



# Gene therapy for sickle cell anemia

